

Are You Ready to Adopt GraphQL?

GraphQL is a query language for APIs and a runtime for fulfilling those queries with your existing data. It is an open source data query language. Developed by Facebook in 2012 for its own use, it was moved to the GraphQL Foundation in 2018.



(Source: <https://www.contentful.com/blog/2018/09/25/3-methods-resolve-graphql-endpoints/>)

In this article, let us explore GraphQL (<https://graphql.org/>), which is a specification from Facebook. GraphQL is a query language for APIs and a runtime for fulfilling those API queries with your existing data. It gives clients the power to ask for exactly what they need and nothing more, by avoiding over-fetching or under-fetching of data. We can understand it more when we see GraphQL implementation in action. Till then, restrain your curiosity. So far we have been using REST (Representation State Transfer) to expose our services as APIs. Let us ask ourselves some questions before going deeper into GraphQL.

Why do I need to adopt GraphQL?

What problems am I facing with REST APIs and how does GraphQL solve them? To answer these queries, let us take a use case of building an e-commerce application for the Web, mobile and native clients. Let's first expose APIs for various e-commerce functionalities. For example, I have the product detail REST API, which gives specific product information as JSON — this includes product data attributes, specifications data, reviews data, etc. As we have many attributes in product JSON, this is larger in size. Each client (Web and thin client —

mobile and tablets) has its own front-end requirements to display product data, as each has different screen sizes, memory, network bandwidth, etc. Now, the clients start consuming the product detail API. Though mobile and tablet

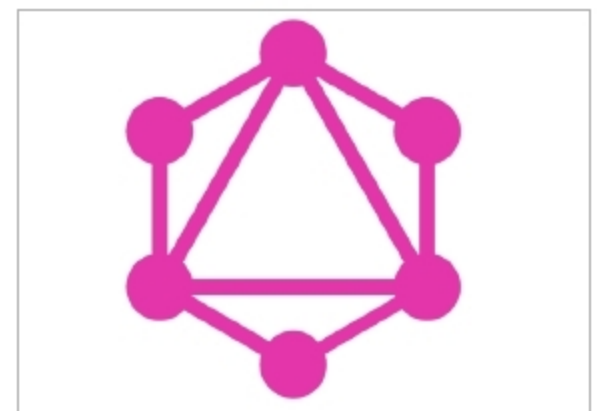


Figure 1: GraphQL logo